FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IX

JEE (Main)-2023

TEST DATE: 28-03-2023

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 questions.

  Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

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Physics

PART – A

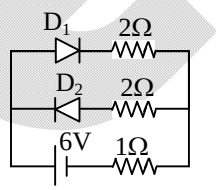
SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

1. The distance covered by a body during the 4th second is twice the distance covered by the body during the 2nd second of this journey. Find the initial velocity of the body if its acceleration is 3 m/s^2.
(A) 2.5 m/s (B) 2.0 m/s
(C) 1.5 m/s (D) 1.0m/s

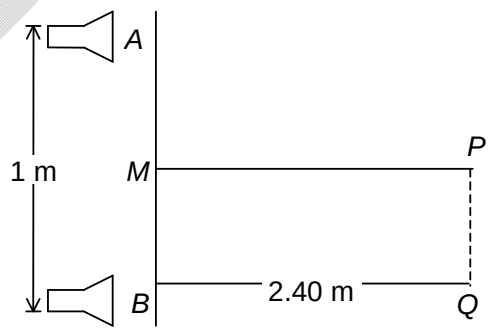
2. Assuming diode to be ideal. The current through the 1Ω resistance in the circuit as shown in the figure is
(A) 2A (B) 1A
(C) 3A (D) none of these



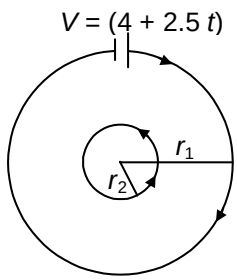
3. A carbon radioactive substance has a half-life time period of 30 days. The disintegration constant is
(A) 231 day^{-1} (B) 231×10^{-2} day^{-1}
(C) 231×10^{-4} day^{-1} (D) 2.31×10^{3} day^{-1}

4. The de Broglie wavelength of 0.08 eV neutron will be (mass of neutron = 1.67×10^{-27} kg)
(A) 1.01×10^{-16} m (B) 1.01×10^{-6} m
(C) 1.01×10^{-10} m (D) 1.01×10^{-12} m

5. Two small loud speakers A, B 1m apart are connected to the same oscillator so that both emit sound waves of a definite frequency in phase. A sensitive detector, moving parallel to AB along PQ, 2.40 m away (see Figure) detects a maxima at P on the perpendicular bisector MP of AB and another maxima when it first reaches a point Q directly opposite to B. The velocity of the sound waves is 340 ms^{-1}. The frequency of the oscillator is
(A) 1500 Hz
(B) 1700 Hz
(C) 3400 Hz
(D) 800 Hz



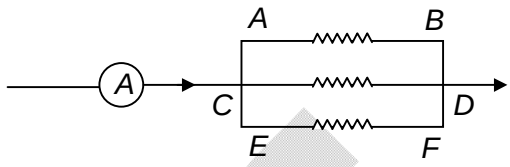
6. Two concentric coplanar circular loops have diameters 20 cm and 2 m and resistance of unit length of each wire is equal to 10^{-4} Ω/m. A time-dependent voltage V = (4 + 2.5 t) volts is applied to the larger as shown. The current in the smaller loop is
(A) 0.5 A (B) 0.75 A
(C) 1 A (D) 1.25 A



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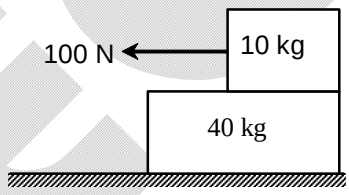
7. A plane mirror is made of glass slab (μ_g = 1.5) 2.5 cm thick and silvered on back. A point object is placed 5 cm in front of the unsilvered face of the mirror. What will be the position of final image?
(A) 12 cm from unsilvered face
(B) 14.6 cm from unsilvered face
(C) 5.67 cm from unsilvered face
(D) 8.33 cm from unsilvered face

8. Figure shows a part of an electric circuit. The wires AB, CD and EF are long and have identical resistances. The separation between consecutive wires is 1 cm. The wires AE and BF have negligible resistance. The ammeter reads 30 amps. The force per unit length on wires AB and CD are



(A) zero, zero
(B) 3×10^{-3} N/m, zero
(C) 2×10^{-3} N/m, 10^{-3} N/m
(D) 10^{-3} N/m, zero

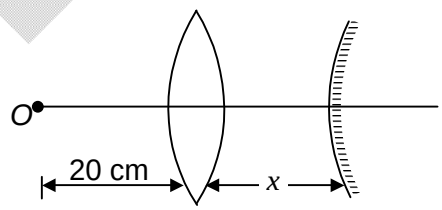
9. A 40 kg slab (B) rests on a smooth floor as shown in figure. A 10 kg block (A) rests on the top of the slab. The static coefficient of friction between slab and block is 0.6 while the kinetic friction coefficient is 0.4. The block (A) is acted upon by a horizontal force 100 N. If g = 9.8 m/s^2, the resulting acceleration of the slab (B) will be



(A) 0.98 m/s^2
(B) 1.47 m/s^2
(C) 1.52 m/s^2
(D) 6.1 m/s^2

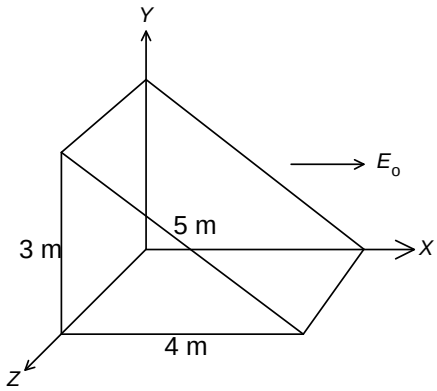
10. Consider telecommunication through optical fibres. Which of the following statements is not true?
(A) Optical fibres may have homogeneous core with a suitable cladding
(B) Optical fibres can be of graded refractive index
(C) Optical fibres are subject to electromagnetic interference from outside
(D) Optical fibres have extremely low transmission loss

11. A point object O is placed at a distance of 20 cm from a convex lens of focal length 10 cm as shown in figure. At what distance x from the lens should a convex mirror of focal length 60 cm, be placed so that final image coincides with the object



(A) 10 cm
(B) 40 cm
(C) 20 cm
(D) final image can never coincide with the object in the given conditions.

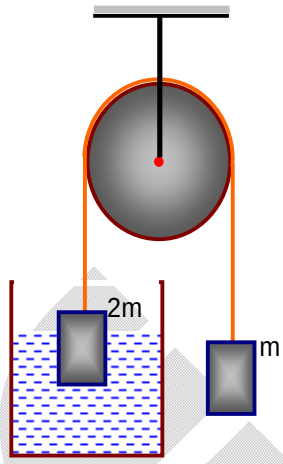
12. The wedge-shaped surface in figure is in the region of uniform electric field E_0 along x axis. The net electric flux for the entire closed surface is



(A) 9 E_0
(B) 15 E_0
(C) 12 E_0
(D) zero

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13. Two homogeneous loads(made of same material) with masses m and 2m are connected by an ideal thread thrown over a fixed pulley, hanging as shown in the figure. Find the density of the material load is made if it is immersed in water by two-thirds of its volume. The density of water is 1 g/cm^3



(A) $\frac{4}{3}$gm / cm^3 (B) $\frac{5}{3}$gm / cm^3

(C) 1.5 gm / cm^3 (D) 2.5 gm / cm^3

14. 70 J of heat is required to raise the temperature of an ideal diatomic gas at constant pressure from 30°C to 35°C. The amount of heat required (in joules) to raise the temperature of the same gas through the same range (30°C to 35°C) at constant volume is

(A) 30 (B) 50

(C) 70 (D) 90

15. A binary star system consists of two stars. One star has twice the mass of the other. The star rotates about their common centre of mass

(A) Star having the smaller mass has twice angular momentum compared to heavier star

(B) Both stars have same angular momentum about centre of mass

(C) Both stars have same linear speed

(D) Both the stars have same kinetic energy

16. A table tennis racket has a flat rectangular handle of mass m and length ℓ and a flat circular disc of radius r and mass m attached to the handle. The moment of inertia of the bat about an axis perpendicular to its plane and passing through its centre of mass is

(A) $\frac{m\ell^2}{6}+\frac{mr^2}{2}+\frac{m}{8}(\ell+2r)^2$ (B) $\frac{m\ell^2}{3}+\frac{3mr^2}{2}$

(C) $\frac{m\ell^2}{12}+\frac{mr^2}{4}+\frac{m}{8}(\ell+2r)^2$ (D) $\frac{m\ell^2}{12}+\frac{mr^2}{2}+\frac{m}{8}(\ell+2r)^2$

17. Two wires of different materials support a massive object suspended from the centre of a uniform horizontal bar with its ends connected to two vertical wires of equal length. What must be the ratio of the diameters of the wires so that the bar may remain horizontal when the object is removed? [Young's modulus of the wires are Y_1 and Y_2 respectively]

(A) $\frac{d_1}{d_2}=\frac{Y_1}{Y_2}$ (B) $\frac{d_1}{d_2}=\frac{Y_2}{Y_1}$

(C) $\frac{d_1}{d_2}=\sqrt{\frac{Y_1}{Y_2}}$ (D) $\frac{d_1}{d_2}=\sqrt{\frac{Y_2}{Y_1}}$

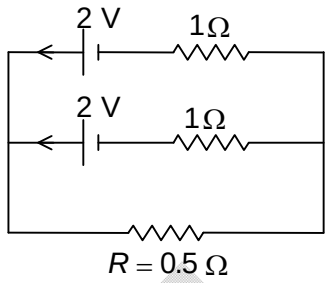
18. A small spherical ball of radius 3.0×10^{-4} m and density 10^4 kg/m^3 falls freely under gravity through a distance h before entering a tank of water. If after entering the water the velocity of the ball does not change, the value of h is. (Viscosity of water is 9.8×10^{-6} N-s/m^2 and density of water is 10^3 kg/m^3)

(A) 1.65×10^3 m (B) 1.65×10^2 m

(C) 1.05×10^3 m (D) 1.05×10^2 m

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19. Two identical batteries each of e.m.f 2 V and internal resistance 1 Ω are connected to a resistance R = 0.5 Ω as shown in circuit. The maximum power that can be developed across R is



(A) 1.28 W (B) $\frac{8}{9}$ W

(C) 2 W (D) 3.2 W

20. The maximum intensity in Young's double slit experiment is I_0. Distance between the slits is d = 5λ, where λ is the wavelength of monochromatic light used in the experiment. What will be the intensity of light in front of one of the slits on a screen at a distance D = 10 d? (Assume d<<D)

(A) $\frac{I_0}{2}$ (B) $\frac{3}{4}I_0$

(C) I_0 (D) $\frac{I_0}{4}$

SECTION – B

(Numerical Answer Type)

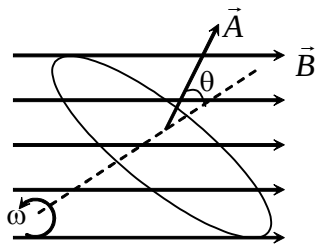
This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

21. A man running at 6 km/hr on a horizontal road in vertically falling rain observes that the rain hits him at 30° from the vertical. If the magnitude of actual velocity of rain is $\mathbf{a}\sqrt{\mathbf{b}}$ km/hr, where a and b are the positive integer ranging between 1 to 9, then find the value of a+b.

22. The wavelength of the first member of the Balmer series in hydrogen spectrum is 6563 Å. What is the wavelength (in Å) of the first member of Lyman series?

23. A longitudinal wave sent by a ship to the bottom of the sea returns after a lapse of $\sqrt{20}$ sec. Elasticity of water is 220 kg/mm^2 and density of sea water is 1.1 gm/cc. The depth of the sea is (in metres)

24. A circular loop of area A = 0.1 m^2 and resistance R = 10Ω rotates with an angular velocity ω = 1 rad/s about an axis through its diameter as shown in figure. The plane of the loop is initially perpendicular to a constant magnetic field B. B = 0.1 T. Find the amplitude of the induced current (in mA) in the loop



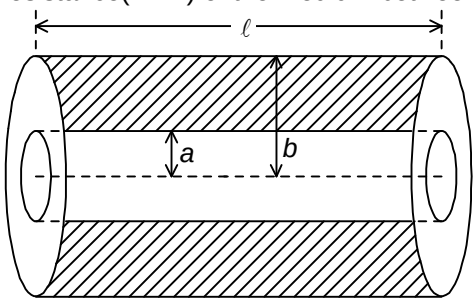
25. A 12 ohm resistance and an inductance of 0.05/π henry with negligible resistance are connected in series. Across the end of this circuit is connected a 130 volt alternating voltage of frequency 50 cycles/second. Calculate the potential difference (in volt) across the inductance.

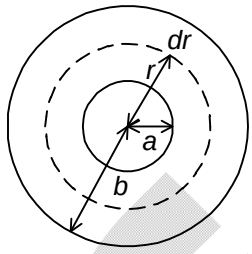
26. In Young's double slit experiment how many maximas can be obtained on a screen (including the central maximum) on both sides of the central fringe if λ = 2000 Å and d = 7000 Å. Given that perpendicular distance of a screen from the mid point of two slits is 3.5 cm

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27. On a slippery road with a coefficient of friction reduced to 0.2, the maximum speed (in m/s) at which a car can go round a curve of radius 100 m is

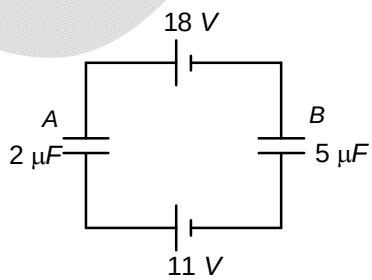
28. A homogeneous poorly conducting medium of resistivity $\rho = 20\ \pi$ SI unit fills up the space between two thin coaxial ideally conducting cylinders. The radii of the cylinders are equal to a and b with a < b (b = 2a), the length of each cylinder is $\ell = 1$ m. Neglecting the edge effects, find the resistance(in Ω) of the medium between the cylinders. ($\log_e 2 = 0.7$)





29. A body of mass m = 1kg falls from a height h = 1m on to the pan of a spring balance. The masses of the pan and spring are negligible. The spring constant of the spring is k= 50N/m. Having stuck to the pan the body starts performing harmonic oscillations in the vertical direction. Find the energy of oscillation in joule.

30. Two capacitors A (2 μF) and B(5 μF) are connected to two batteries as shown in the figure. Then the potential difference in volts between the plates of A is



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Chemistry

PART – B

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

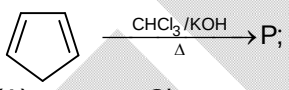
31. Which of the following is correct?
(A) If $AgNO_3\,(aq)$ is added to a dilute KI solution, then positively charged sol is formed.
(B) If $FeCl_3$ is mixed with cold water, negatively charged sol of $Fe(OH)_3$ is formed.
(C) If $FeCl_3$ is added to dil. NaOH solution, negatively charged sol of $Fe(OH)_3$ is formed.
(D) Gold sol is a lyophilic sol.

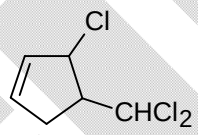
32. Which of the following transition is forbidden electron transition?
(A) $3s \rightarrow 2s$ (B) $3p \rightarrow 2s$
(C) $4d \rightarrow 3p$ (D) $4p \rightarrow 1s$

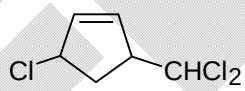
33. Which of the following is a pair of isoelectronic as well as isostructural species?
(A) NO_3^-, CO_3^{2-} (B) CO_2, $\overset{\oplus}{N}O_2$
(C) XeF_2, IF_2^- (D) All of these

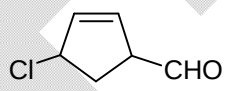
34. Which of the following is correct for an ideal gas?
(A) Fraction of molecules with too high velocity is large enough at high temperature.
(B) Fraction of molecules with too low velocity is very small at low temperature.
(C) On increasing temperature the fraction of molecules attaining high velocity increases more rapidly.
(D) On increasing temperature the fraction of molecules attaining most probable speed increases.

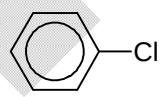
35. What is the product 'P' of the following reaction?



(A) 

(B) 

(C) 

(D) 

36. Which of the nitrates on strong heating leaves a metal as the residue?
(A) $Hg(NO_3)_2$ (B) $Pb(NO_3)_2$
(C) $Cu(NO_3)_2$ (D) $Ca(NO_3)_2$

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37. Fe^{3+} solution forms a blue coloured precipitate on mixing with $K_4\left[Fe(CN)_6\right]$. What is the compound?

(A) $Fe_3\left[Fe(CN)_6\right]_6$ (B) $Fe_4\left[Fe(CN)_6\right]_3$

(C) $Fe\left[Fe(CN)_6\right]$ (D) $Fe_3\left[Fe(CN)_6\right]_2$

38. Which of the following impart violet colour to their aqueous solution?

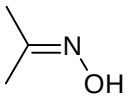
(A) $\left[Cr(H_2O)_6\right]^{3+}$ (B) $\left[Cr(H_2O)_4Cl_2\right]^{+}$

(C) $\left[Fe(CN)_6\right]^{4-}$ (D) CrO_4^{2-}

39. Which of the following is most acidic?

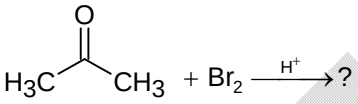
(A) $NH_2-\overset{\oplus}{N}H_3$

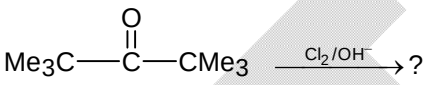
(B) NH_2OH

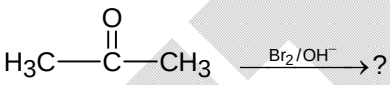
(C) 

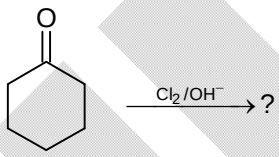
(D) CH_3COOH

40. In which of the following process haloform formation is favourable?

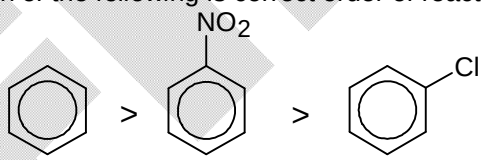
(A) 

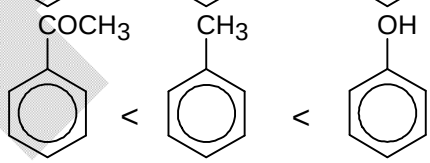
(B) 

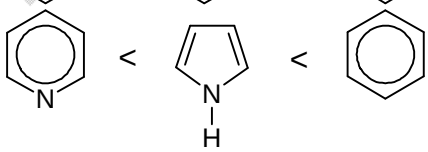
(C) 

(D) 

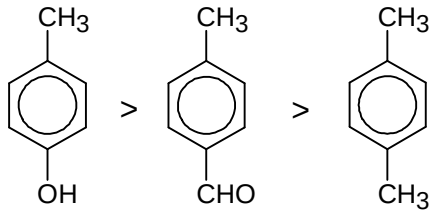
41. Which of the following is correct order of reactivity towards EAS?

(A) 

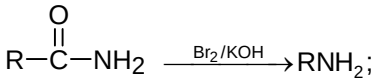
(B) 

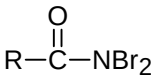
(C) 

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(D) 

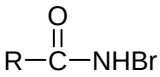
42. Which of the following is involved as intermediate in the following reaction?



(A) 

(B) RNHBr

(C) R – C = N = O

(D) 

43. Given

$E^{o}_{Na^{+}/Na} = -2.71\ V$; $E^{o}_{H_2O/H_2} = -0.83\ V$; $E^{o}_{Cl^{-}/Cl_2} = -1.36\ V$; $E^{o}_{H_2O/O_2} = -1.23\ V$

Which of the following is correct for electrolysis of aq. NaCl?

(A) Cl_2 gas is liberated at anode.

(B) O_2 gas is liberated at anode.

(C) Na(s) is deposited at cathode.

(D) pH of the solution decreases.

44. Which of the following process is used for purification of Nickel?

(A) Cupellation (B) Mond's process

(C) van Arkel process (D) Zone refining

45. In which of the following, C – O bond length is longest?

(A) $\left[Mn(CO)_6\right]^{2+}$ (B) $\left[Fe(CO)_5\right]$

(C) $\left[Cr(CO)_6\right]$ (D) $\left[V(CO)_6\right]^{-}$

46. Which of the following is a chiral analgesic?

(A) Dichlofenac (B) Ibuprofen

(C) Aspirin (D) Ranitidine

47. Which of the following combination of method of preparation, raw material and product is correct?

(A) Down's process, electrolysis of fused NaOH, Na

(B) Castner process, electrolysis of fused NaOH, Na

(C) Nelson process, double decomposition of Na_2CO_3,NaOH

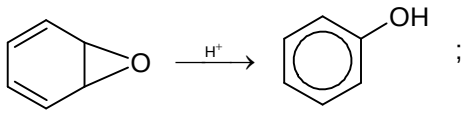
(D) Gossage process, electrolysis of aq. NaCl, NaOH

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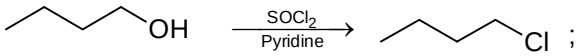
48. In which of the following process, a carbocation is involved as an intermediate?

(A) $C_2H_5 - O - C_2H_5 \xrightleftharpoons{HBr} C_2H_5OH + C_2H_5Br$;

(B) $CH_3CH_2CH = CH_2 \underset{HBr}{\overset{ROOR}{\rightleftharpoons}} CH_3CH_2CH_2CH_2Br$;

(C) 

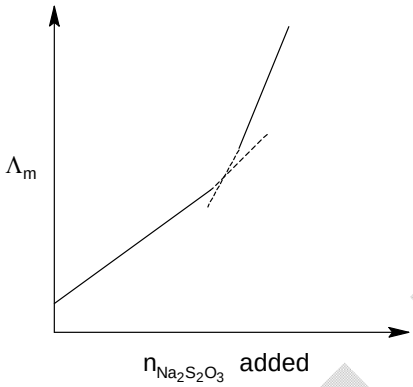
;

(D) 

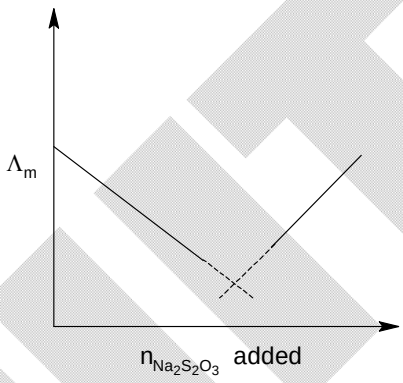
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49. Which of the following is correct conducometric titration curve?

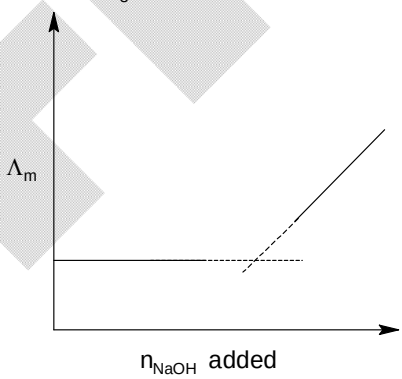
(A) AgCl vs. $Na_2S_2O_3$



(B) AgCl vs. $Na_2S_2O_3$

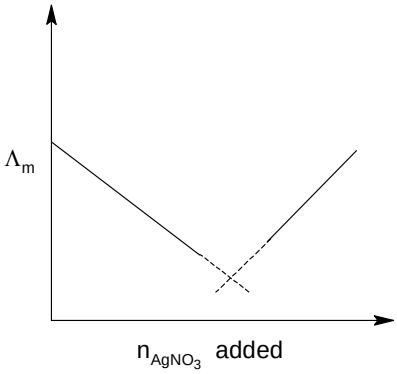


(C) CH_3COOH v / s NaOH

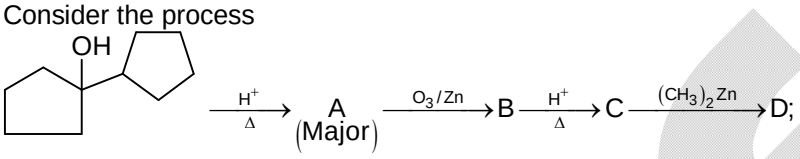


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(D) KCl vs. $AgNO_3$



50. Consider the process



Which of the following is correct with reference to the given process?
(A) B exhibits geometrical isomerism.
(B) A can give Tollen's test.
(C) D exhibits optical isomerism.
(D) Degree of unsaturation of 'D' is 5.

SECTION – B

(Numerical Answer Type)

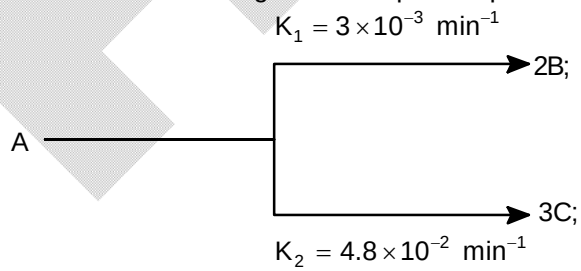
This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. Aqueous solution of perbenzoic acid $(pK_a = 4.0)$ is titrated against aq. NaOH solution. What is its pH at 50% equivalence point?

52. The following are the thermodynamic variables (with their usual meaning). How many of these are state function(s) or state variable(s)?
G, P, H, S, U, V, A, T, Q, W.

53. In NaCl lattice, Cl^- constitute CCP lattice while Na^+ ions occupy all the octahedral voids. What is the number of third nearest neighbours to Cl^- ions?

54. Consider the following first order parallel process:



What is the mole% of B in product mixture, at the end of the process?

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55. 0.1 mole NH_3 gas is passed into 1 litre of a saturated solution of AgOH. Finally, the Ag^+ ion concentration is found to be 7.434×10^{-x} M. What is the value of x? Given

$K_{SP}\left(AgOH\right) = 2 \times 10^{-8}\ M^2$, $K_f = \left\{\left[Ag\left(NH_3\right)_2\right]^{2+}\right\} = 1.7 \times 10^{+7}$

56. How many of the following are planar molecules (having all the constituent atoms in same plane)?

H_2O, NH_3, H_2O_2, I_3^-, XeF_4, SF_4, $H_2C = C = CH_2$, B_2H_6, Be_2H_4, $SOCl_2$, SCl_2, XeF_2, I_2Cl_6.

57. During metallurgical extraction, how many of the following are extracted involving carbon reduction?

Pb, Mn, Cr, Mg, Al, Hg, Fe, Zn, Sn, Cu.

58. How many of the following are soluble in excess of NaOH solution?

$Al(OH)_3$, $Fe(OH)_3$, $Be(OH)_2$, $Sb(OH)_3$, Bi_2S_3, ZnS, $PbCrO_4$, Ag_2S, $PbSO_4$.

59. 0.8 gm of an organic compound give 0.1692 gm of a yellow compound on heating with fuming nitric acid and subsequent addition of a few crystals of silver nitrate in the same tube. What is the percentage of 'Br' in the organic compound?

60. How many of the following electronic transitions is/are permissible transitions in a hydrogen atom?

(a) $3p \rightarrow 2s$ (b) $3s \rightarrow 2p_z$ (c) $4p_z \rightarrow 3d_{z^2}$ (d) $4p_z \rightarrow 3d_{xy}$ (e) $4p_z \rightarrow 3s$

(f) $3d \rightarrow 1s$ (g) $4d \rightarrow 3d$ (h) $4p \rightarrow 3p$ (i) $3p_z \rightarrow 2p_x$ (j) $2s \rightarrow 1s$

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Mathematics

PART – C

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

61. Let A = {1, 2, 3, 4, 5, 6, 7, 8, 9} and B = {2, 5, 8, 11}, then the number of elements in the set $\{C \subseteq A : C \cap B \neq \phi\}$ is
(A) 8 (B) 384
(C) 448 (D) 480

62. For $z \in C$ if the minimum value of $\left(\left|z-4\sqrt{3}\right|+\left|z-k\sqrt{3}i\right|\right)$ is $5\sqrt{3}$, then a value of k is
(A) 3 (B) $\sqrt{3}$
(C) 2 (D) 4

63. If the coefficient of x^6y^4 in the expansion of $(x + 2y + 3xy)^9$ is $k \times 2^6 \times 3^3$, then k is equal to
(A) 21 (B) 7
(C) 5 (D) 3

64. If $\frac{1}{(15-k)(30-k)}+\frac{1}{(30-k)(45-k)}+\frac{1}{(45-k)(60-k)}+.....$
$+\frac{1}{(285-k)(300-k)}+\frac{1}{(300-k)(315-k)}=\frac{1}{155}$, then maximum value of k is
(A) 25 (B) 325
(C) 400 (D) 335

65. Let L_1 be a straight line passing through origin and L_2 be the line x + y = 1. If the intercepts made by the circle $x^2 + y^2 - x + 3y = 0$ on L_1 and L_2 are equal, then which of the following can represent L_1
(A) x + y = 0 (B) x – y + 1 = 0
(C) x + 7y = 0 (D) x – 7y = 0

66. If the chord of contact of pair of tangents from a point P on $\frac{x^2}{9}+\frac{y^2}{16}=1$ to the ellipse $\frac{x^2}{3}+\frac{y^2}{4}=1$ is normal to the curve $2x^2 + 2y^2 - 4x - 4y + 1 = 0$, then eccentric angle of point P can be
(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$
(C) $\frac{\pi}{2}$ (D) $\frac{\pi}{6}$

67. Tangents are drawn from the point $(\alpha, 2)$ to different branches of the hyperbola $x^2 - 4y^2 - 2x - 16y - 31 = 0$, then α lies in the interval
(A) (–4, 4) (B) (–1, 1)
(C) (–5, 7) (D) (–7, 9)

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68. If the vertex of the parabola is (1, –1) and equation of its directrix is 3x – 4y + 3 = 0, then length of latus rectum is

(A) $\frac{8}{5}$ (B) 4

(C) 8 (D) 2

69. A random variable X has the probability distribution

X	1	2	3	4	5	6	7	8	9	10	11
P(X)	0.12	0.05	0.07	0.10	0.08	0.03	0.20	0.15	0.06	0.09	0.05

For the events E = {X is an odd prime number} and F = {X ≤ 6}, the P(E∪F) is

(A) 0.85 (B) 0.65

(C) 0.70 (D) 0.15

70. If the image of the line $L_1 : \frac{x-1}{2} = y = \frac{z+1}{2}$ in the plane P : x – 4y + z = 9 is L_2 then L_2 passes through which of the following point :

(A) (7, 3, 5) (B) (2, 4, 2)

(C) (4, –3, 2) (D) (0, 3, –3)

71. Let $\vec{a},\vec{b},\vec{c}$ be 3 non-zero vectors such that $\vec{a}\cdot\vec{b}=1, \vec{b}\cdot\vec{c}=-2, \vec{c}\cdot\vec{a}=4$ and the volume of the tetrahedron whose coterminous edges are $\vec{a},\vec{b},\vec{c}$ is $\frac{16}{3}$, then the value of $\left[\vec{a}\times\left(\vec{b}\times\vec{c}\right),\vec{b}\times\left(\vec{c}\times\vec{a}\right),\vec{c}\left(\vec{b}\times\vec{a}\right)\right]$ is equal to

(A) 0 (B) 512

(C) –512 (D) $\frac{256}{3}$

72. If $\alpha, \beta, \gamma, \delta$ are the roots of the equation $x^4 - 5x^2 + 2x + 3 = 0$, then the value of $(\alpha+\beta+\gamma)^{-1} + (\alpha+\beta+\delta)^{-1} + (\alpha+\gamma+\delta)^{-1} + (\beta+\gamma+\delta)^{-1}$ is equal to

(A) $\frac{2}{3}$ (B) $-\frac{2}{3}$

(C) $\frac{3}{2}$ (D) $\frac{5}{3}$

73. The number of 6 digit natural numbers, such that product of their digits is 36 is

(A) 300 (B) 360

(C) 405 (D) 375

74. Let $A = \begin{bmatrix} 1 & 0 & 0 \\ k & 1 & 0 \\ k & m & 1 \end{bmatrix}$, k, m ∈ R. If for some n ∈ N $A^n = \begin{bmatrix} 1 & 0 & 0 \\ 16 & 1 & 0 \\ 184 & 24 & 1 \end{bmatrix}$, then n + k + m is equal to

(A) 16 (B) 13

(C) 22 (D) 10

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75. Let k be an integer such that $\lim_{x\to 6}\frac{15-[2-x]}{[x-2k]}$ exists, where [t] is greater integer $\le$ t. Then k is equal to

(A) –7 (B) 7
(C) 3 (D) –3

76. If $\theta=\frac{\pi}{8}$ and A = $\begin{bmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{bmatrix}$ and if B = $A^2 + A^6$, then det(B) is equal to

(A) –2 (B) $2\sqrt{2}$
(C) 2 (D) $\frac{1}{2}$

77. Let m denote the number that turns up when a fair die is rolled. If the probability that the system of equations x – 2y + 3z = 6, my + z = 2 and 3x – 24y + mz = 3 has unique solution is p, then sum of all possible values of m and 3p

(A) 14 (B) 13
(C) 15 (D) 16

78. $\cot^{-1}\left(\frac{3+\sqrt{3}}{1+\sqrt{3}}\right)+\cos^{-1}\left(\sqrt{\frac{6+3\sqrt{3}}{8+4\sqrt{3}}}\right)$ is equal to

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$
(C) $\frac{\pi}{3}$ (D) $\frac{\pi}{6}$

79. Let f be a differentiable function defined on $\left[0,\frac{\pi}{3}\right]$ such that f(x) > 0 and

$f(x)+\int_0^x f(t)\sqrt{1-\left(\log_e\left(f(t)\right)\right)^2}\,dt=\sqrt{e^{\sqrt{3}}}\quad\forall\, x\in\left[0,\frac{\pi}{3}\right]$, then $\ln\left(f\left(\frac{\pi}{4}\right)\right)$ is equal to

(A) $\frac{\sqrt{3}-1}{2\sqrt{2}}$ (B) $\frac{1}{2}$
(C) $\frac{1}{\sqrt{2}}$ (D) $\frac{1}{2\sqrt{2}}$

80. ~ q $\rightarrow$ (~ p $\Leftrightarrow$ q) is false, then the truth values of p and q are respectively

(A) T, T (B) T, F
(C) F, T (D) F, F

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. If the area of the region A = {(x, y) : $x^2 + y^2 \le 9$, $x^2 + 4y^2 \ge 4$ and y $\le$ 0} is $\frac{k\pi}{2}$, then k is equal to

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82. Let $f(t)=\int_0^t e^{x^4}\left(\frac{x^{15}}{\left(x^{12}+3x^8+6x^4+1\right)^2}\right)dx$. If $f(1)-f'(1)=ke-\frac{1}{4}$, then 484k is equal to

83. Let $f: R \to R$ be a differentiable function such that $f(x+y)=f(x)+f(y)+2xy-3\ \forall\ x, y \in R$. If $f'(0)=-1$, then f(3) is equal to

84. If the planes $\vec{r}\left(\hat{i}+\hat{j}+\hat{k}\right)=p$, $\vec{r}\left(\hat{i}+2a\hat{j}+\hat{k}\right)=p$ and $\vec{r}\left(a\hat{i}+a^2\hat{j}+\hat{k}\right)=p$ intersect in a line, then integral value of a is

85. If $\sin2°+\sin4°+\sin6°+\ldots\ldots\sin178°=\cot k°$, then k is equal to

86. Let k be an integer for which both root of the quadratic equation $x^2+2kx+9=6x$ lies in $(-6, 1)$. If $2, a_1, a_2, a_3, \ldots\ldots a_{19}, a_{20}, k$ are in G.P., then the value of $\frac{a_4a_{17}}{2}$ is

87. If $\sum_{k=1}^{2025}\tan\left(\frac{\alpha}{2^k}\right)\sec\left(\frac{\alpha}{2^{k-1}}\right)=\tan\left(\frac{\alpha}{2^p}\right)-\tan\left(\frac{\alpha}{2^m}\right)$, then $\frac{p+m}{225}$ is equal to

88. $\sum_{k=1}^{25}\left(k^2\left({}^{25}C_k-{}^{25}C_{k-1}\right)+(2k+1)\,{}^{25}C_k\right)=A$, then $\frac{A}{135}$ is equal to

89. An equilateral triangle has each of its sides of length $\sqrt{2}$ cm. If (x_1, y_1), (x_2, y_2) and (x_3, y_3) are the vertices, then the value of determinant $\begin{vmatrix} x_1^2+x_2^2+x_3^2 & x_1y_1+x_2y_2+x_3y_3 & x_1+x_2+x_3 \\ x_1y_1+x_2y_2+x_3y_3 & y_1^2+y_2^2+y_3^2 & y_1+y_2+y_3 \\ x_1+x_2+x_3 & y_1+y_2+y_3 & 3 \end{vmatrix}$ is equal to

90. The remainder when 3^{140} is divided by 12 is

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